

Integrated Pandas Practical 1

Retail Store Sales Analysis

Student Assignment - No Answers

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Scenario: You are a junior data analyst at a retail company. The manager has received a monthly sales file, but the data may contain missing values and duplicate transactions. Your job is to inspect, clean, and analyze the data before reporting useful business findings.

Chapters covered: Chapter 1 Sorting, Chapter 2 Grouping, Chapter 3 Missing Values, and Chapter 4 Duplicate Handling.

Dataset: retail_sales_practical.csv

Student instruction: Complete the questions in order. This file intentionally contains no answers. Show your Pandas code, output, and short interpretation wherever requested.

1. Business Dataset

```
import pandas as pd
```

```
df = pd.read_csv("retail_sales_practical.csv")
```

The first 10 rows are shown below only to help you understand the structure. Use the complete CSV for your work.

Order_ID	Date	Product	Category	City	Quantity	Price	Salesperson
O101	01-07-2026	Laptop	Electronics	Surat	1	52000	Aarav
O102	01-07-2026	Mouse	Electronics	Mumbai	3	700	Diya
O103	02-07-2026	Chair	Furniture	Surat	2	4500	Rohan
O104	02-07-2026	Keyboard	Electronics	Ahmedabad		1500	Priya
O105	03-07-2026	Desk	Furniture	Surat	1		Aarav
O106	03-07-2026	Monitor	Electronics	Mumbai	2	12500	Diya
O107	04-07-2026	Notebook	Stationery	Ahmedabad	10	80	Rohan
O108	04-07-2026	Pen Set	Stationery	Surat	5	250	Priya
O109	05-07-2026	Office Chair	Furniture	Mumbai	1	7800	Aarav
O110	05-07-2026	Printer	Electronics	Ahmedabad	1	16500	Diya

2. Part A - Initial Inspection

1. Load the CSV into a DataFrame named df.
2. Display the first five rows.
3. Find the number of rows and columns.

4. Display all column names.
5. Inspect column data types and non-null counts.
6. Write two observations about possible data-quality problems.

3. Part B - Missing Values

1. Count missing values in each column.
2. Display only rows that contain at least one missing value.
3. Find the total number of missing cells.
4. Treat the missing Quantity using a sensible summary value and explain why.
5. The missing Price belongs to a Furniture item. Fill it using the median Furniture price.
6. Verify that no missing values remain.

4. Part C - Duplicate Handling

1. Count later exact duplicate rows.
2. Display all rows involved in exact duplicate groups.
3. Check repeated Order_ID values using subset.
4. Explain why a repeated Product, City, or Salesperson does not automatically mean a duplicate transaction.
5. Remove confirmed exact duplicates and reset the index.
6. Verify that no exact duplicates remain.
7. Report the final number of cleaned transactions.

5. Part D - Create a Sales Measure

1. Create a new column $\text{Sales} = \text{Quantity} \times \text{Price}$.
2. Display Order_ID, Product, Quantity, Price, and Sales for the first five cleaned rows.
3. Explain why creating Sales is useful for the later analysis.

6. Part E - Sorting

1. Sort all cleaned transactions by Sales from highest to lowest.
2. Display the top five transactions by Sales.
3. Display the three lowest-value transactions.
4. Sort first by City and then by Sales from highest to lowest within each city.

5. Find the highest-value transaction in the cleaned dataset.

7. Part F - Grouping

1. Calculate total Sales by City.
2. Calculate total Sales by Category.
3. Count transactions handled by each Salesperson.
4. Calculate total Sales by Salesperson.
5. Calculate total Quantity sold by Category.
6. Calculate average Sales by City.
7. Group by both City and Category and calculate total Sales.
8. For each Category, calculate count, sum, mean, minimum, and maximum Sales.

8. Part G - Management Questions

1. Which city generated the highest total Sales?
2. Which category generated the highest total Sales?
3. Which salesperson generated the highest total Sales?
4. What is the total Sales value of the cleaned dataset?
5. Which product has the highest single transaction Sales?
6. Display all transactions with Sales greater than 10,000, sorted from highest to lowest.
7. What business conclusion can you give the manager after reviewing city, category, and salesperson performance?

9. Submission Checklist

- ☐ All questions attempted in sequence.
- ☐ Code and outputs included.
- ☐ Missing values inspected before treatment.
- ☐ Duplicates inspected before removal.
- ☐ Cleaning verified before analysis.
- ☐ Sorting and grouping questions completed.
- ☐ Short business interpretations added.